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logical division of ideas

Limitations of quantum computing

In the first half of the 20th century the brightest physicist of the time revolutionized their field by introducing quantum mechanics. After its principles were translated into mathematical language, it was adopted by computer scientists, giving birth to quantum computing. By taking advantage of quantum superposition and entanglement, it allows for operations impossible on classical computers. Quantum computing is a new and exciting field that could potentially provide answers to the most difficult computational problems, such as large number factorization and calculating discrete logarithms. Despite producing promising results, that goal remains elusive. Quantum computing continues to be plagued with numerous **issues** that hold back its progress.

There are multiple ways of constructing quantum computers, but all of them are complicated and costly. Classical computers use electrical voltage to represent the value of a bit – state 0 or 1. However a quantum bit, called qubit, can be in coherent superposition of both states. Only when measured it collapses to 0 or 1 with some probability. Many microscopic particles have these characteristics, for example photons, semiconductor nanocrystals or singular atoms. To use them though, the apparatus that can isolate and measure such small particles has to be very precise. Rare, expensive material are often needed to construct it. Additionally, very low temperature has to be maintained inside the machine, resulting in high energy use. Thus, only big and highly technologically advanced companies can afford to build quantum computers.

Even in perfect conditions, quantum circuits are very error-prone. The most sophisticated existing quantum computers cannot always maintain superposition state. If superposition of a qubit is lost, its measured value may be different than expected. Due to the nature of the uncertainty principle, it is impossible to avoid all such errors. External interference, mainly mechanical vibrations, fluctuations of voltage from a power supply or temperature changes, cause additional hardware errors. The combination of occurring errors creates noise that corrupts the output of quantum computations. Improving machinery and increasing the number of qubits can mitigate errors, but it is a slow process. Thus, the results of quantum algorithms cannot be fully trusted. Whilst it is possible to use error correcting codes, they increase the size of input and inhibit the computations.

The size of input that quantum computers operate on is too small to be useful. Currently, quantum algorithms can only be performed on small-sized data. A quantum computer may be able to factorize a two digit number faster than any classical computer, but the numbers commonly used in cryptography have thousands of digits. To be applicable commercially, it is imperative that quantum computers can process large data. A major obstacle to this is the insufficient number of qubits. Although newer quantum computers have more qubits available, the rate at which their number grows is not satisfactory. For now, classical computers can operate on inputs larger by orders of magnitude. It is possible that quantum computers will always be inferior in this regard.

There are few computational problems for which quantum algorithms exist. Shor's and Grover's algorithms provide time-efficient solutions to the discrete logarithm and database search problems. In the future they may be crucially important in cryptography and information technology. However most other quantum algorithms, for example Deutsch–Jozsa or Simon's algorithms, are not practical or useful. They were invented by scientists to prove that the quantum technology is worth exploring. Constructing algorithms that can be used in business is more difficult. There is no way of knowing if a given problem is possible to be solved on a quantum computer, and if the solution will be efficient. Quantum computing could be constrained to a small set of specialized applications and stagnate.

Quantum computing is still a very new field of science. In time, all of the above problems it now faces might be solved. Increased funding could enable construction of more powerful quantum computers that will be able to perform computations on larger input data and generate less errors. Devoted researchers may find ways of simplifying the design of quantum computers and invent new efficient quantum algorithms. The current limits of quantum computing can be overcome.